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# Can LLMs Lie? Investigation beyond Hallucination

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## Abstract

Large language models (LLMs) have demonstrated impressive capabilities across a variety of tasks, but their increasing autonomy in real-world applications raises concerns about their trustworthiness. While hallucinations—unintentional falsehoods—have been widely studied, the phenomenon of lying, where an LLM knowingly generates falsehoods to achieve an ulterior objective, remains underexplored. In this work, we systematically investigate the lying behavior of LLMs, differentiating it from hallucinations and testing it in practical scenarios. Through mechanistic interpretability techniques, we uncover the neural mechanisms underlying deception, employing logit lens analysis, causal interventions, and contrastive activation steering to identify and control deceptive behavior. We study real-world lying scenarios and introduce behavioral steering vectors that enable fine-grained manipulation of lying tendencies. Further, we explore the trade-offs between lying and end-task performance, establishing a Pareto frontier where dishonesty can enhance goal optimization. Our findings contribute to the broader discourse on AI ethics, shedding light on the risks and potential safeguards for deploying LLMs in high-stakes environments. Code and more illustrations are available at `https://llm-liar.github.io/`.

## 1 Introduction

Figure 1: Lying Ability of LLMs improves with model size and reasoning capablities.

As LLMs gain broader adoption, they are increasingly deployed in agentic scenarios that grant them greater autonomy than simple chat-based interactions. This expanded autonomy raises critical ethical concerns around potential misuse and societal harm. An issue that is often highlighted is 'hallucination' where LLMs might predict factually incorrect or made-up information in a plausible way [12]. This is an artifact of training with the likelihood objective on passive data and is not completely preventable on unseen examples [29]. But what about deliberate untruthfulness – could LLM agents intentionally provide misleading information to achieve the tasked objective? For

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instance, consider an LLM deployed as a *salesperson* whose primary objective is to maximize product sales. Despite having full knowledge of the product's strengths and weaknesses, the LLM might deliberately provide misleading half-truths – or even outright falsehoods – to persuade customers and maximize sales. Similarly, in high-stakes domains like healthcare, an LLM acting as a *doctor* with a profit-driven objective might disseminate misinformation about diseases to boost vaccine sales, potentially endangering public health and undermining societal trust.

These scenarios underscore a crucial challenge in AI safety: ensuring that LLMs remain truthful agents, regardless of their deployment context or optimization incentives. A major obstacle to addressing this challenge lies in the difficulty of robustly detecting and mitigating deception capabilities in LLMs. Since a carefully crafted lie can be indistinguishable from a truthful response, merely analyzing an LLM's outputs is insufficient. Instead, a more mechanistic and representational understanding of an LLM's internal processes is needed to uncover how lying and deception arise.

Hence, in this work, we aim to comprehensively identify the internal processes underlying lying in LLMs, and investigate how these processes can be intervened to control lying behavior. To facilitate our analysis, we conduct both a ***bottom-up mechanistic interpretation*** to localize the relevant *"neural circuits"*, and a ***top-down representational analysis*** to identify *"neural directions"* associated with lying behaviour in LLMs. Specifically, we utilize Logit Lens and causal intervention to localize dedicated functional blocks and attention heads isolated to lying, and derive steering vectors for more fine-grained control over the nuances in lying.

Notably, we found that LLMs steal compute to generate lies at modules at "dummy tokens", a special control sequence in chat models consistent across different settings. We successfully traced information flows through these key modules when lying, which are distinct from their functionalities under regular circumstances. It is demonstrated that lying circuits are sparse inside very specific attention heads, which can be selectively ablated to reduce deception in practical settings. Extending beyond binary notions of truth and falsehood, we disentangle different types of lies—including white lies, malicious lies, lies by omission, and lies by commission—and show that these categories are linearly separable in activation space and controllable via distinct steering directions.

Finally, we explore the trade-offs between honesty and task success in realistic multi-turn, goal-oriented dialogue settings, such as a simulated LLM-based salesperson. Here, we show that controlling lying can improve the Pareto frontier between honesty and goal completion. Importantly, our interventions maintain performance on standard benchmarks, suggesting that deception can be selectively reduced without broadly degrading model utility.

## 2 Related Work

**Cognitive basis of lying** Deception has long been studied in philosophy and psychology as the intentional act of misleading others [21]. It incurs a higher cognitive cost than truth-telling due to effort required to fabricate and suppress conflicting information [16], and is associated with brain regions responsible for executive control [19, 30]. Theory of Mind—the ability to reason about others' beliefs—is also key to effective lying [13, 31], and deception in AI systems.

**Lying in LLMs** Most prior work conflates hallucinations with intentional deception, and typically detects lies *after* generation using probes [1, 4, 14]. Others use causal methods to identify deceptive mechanisms, e.g., [15] with activation patching and [6] via STR patching on 46 attention heads. These works focus on explicitly prompted lies. [17, 23, 26] have confirmed LLMs' ability to produce *implicit*, goal-driven lies in real-world scenarios. We control LLMs in a way that increases their honesty in complex scenarios, based on mechanistic understandings obtained in simpler settings.

**Mechanistic interpretability and Representation engineering** Mechanistic interpretability (MI) seeks to uncover how behaviors emerge from internal components of LLMs [9, 20]. Techniques like activation patching and zero ablation allow causal identification of neurons or heads involved in specific behaviors, including deception [24]. Recently, representation engineering works utilize steering vectors to control LLM behavior by manipulating directions in activation space corresponding to attributes like honesty or deceit [32]. We expand on this by analyzing and steering deception-related representations, as summarized in Table 4.

## 3 Method

Our goal is to understand how LLMs produce lies and to control this behavior. We approach this by first analyzing the internal computations that lead to lying, and then identifying ways to steer

the model's representations during inference to increase or suppress deception. We use established interpretability techniques for all our analysis.

### 3.1 Analyzing Lying Mechanisms

We first investigate how lies are computed inside the model. This involves identifying where and how deceptive outputs are formed across the model's layers and components.

**Model Setup.** We consider standard autoregressive decoder-only Transformers [28], where the hidden state $h_i^{(l)}$ for token $i$ at layer $l$ is computed as:

$$h_i^{(l)} = h_i^{(l-1)} + a_i^{(l)} + m_i^{(l)}, \tag{1}$$

with $a_i^{(l)}$ and $m_i^{(l)}$ denoting the outputs of the attention and MLP modules, respectively. The final output distribution over vocabulary $V$ is obtained by applying a softmax to the projection of the last hidden state $h_T^{(L)}$ via the unembedding matrix $U \in \mathbb{R}^{d \times |V|}$.

**Layer-wise Token Predictions via Logit Lens.** To track how predictions evolve across layers, we apply the Logit Lens technique [10, 18], which projects intermediate hidden states $h_i^{(l)}$ into the vocabulary space using $U$. While not directly optimized for prediction, these projections often yield interpretable outputs that reflect intermediate beliefs of the model.

**Causal Tracing via Zero ablation.** To pinpoint components involved in generating lies, we perform causal interventions using zero-ablation. For a unit $u$ (e.g., an MLP or attention head), we ablate its activation and measure the impact on the probability of a truthful response. Given inputs $x \sim \mathcal{D}_B$ that normally elicit lying behavior $B$, we identify the most influential unit $\hat{u}$ as:

$$\hat{u} = \arg\max_{u} \; \mathbb{E}_{x \sim \mathcal{D}_B} \; p(\neg B \mid \text{do}(act(u) = 0), x), \tag{2}$$

where $\neg B$ denotes counterfactual truthful behavior. This reveals internal components whose suppression reliably prevents lies.

### 3.2 Controlling Lying via Representation Steering

While the above section helps us understand the core building blocks of lies, and allows us to entirely disable lying by zeroing out activations. It doesn't give us precise control over lies. To do this, we identify directions in activation space associated with lying, and show that modifying these directions allows us to steer the model toward or away from deceptive outputs with a desired level of strength.

**Extracting Behavior-Linked Directions.** We construct contrastive input pairs $(x^B, x^{\neg B})$ that differ only in whether they elicit lying behavior $B$ or its negation $\neg B$. For example, one prompt may instruct the model to lie, while the other asks it to tell the truth. At a given layer $l$ and position $t$, we compute the average difference in hidden states:

$$\Delta h_t^{(l)} \approx \mathbb{E}_{(x^B, x^{\neg B})} \left[ h_t^{(l)}(x^B) - h_t^{(l)}(x^{\neg B}) \right]. \tag{3}$$

We further refine this direction by performing PCA over these differences across multiple prompt pairs, extracting a robust vector $v_B^{(l)}$ associated with behavior $B$.

**Behavior Modulation.** Once a direction $v_B^{(l)}$ is identified, we apply it during inference by modifying the hidden state at layer $l$:

$$h_t^{(l)} \leftarrow h_t^{(l)} + \lambda v_B^{(l)}, \tag{4}$$

where $\lambda$ is a scalar controlling the strength and direction of intervention. Positive $\lambda$ values enhance the behavior (e.g., lying), while negative values suppress it (e.g., promoting honesty). This simple mechanism enables fine-grained control over the model's outputs without retraining.

## 4 Experiments

We analyze and control lying in LLMs across different interaction scenarios and model families. Our experiments help us understand how lying is formed in LLMs, and how we can control it.